

Towards a mathematical theory of science

1 Overview

We propose a formal framework for representing scientific discourse as trajectories over a semantic embedding space. The framework distinguishes carefully between:

- (i) empirical objects constructed directly from data,
- (ii) statistical structures inferred from empirical regularities,
- (iii) and theoretical interpretations imposed on top of those structures.

The framework does not assume that embeddings directly represent the true geometry of knowledge. Instead, embeddings are treated as noisy coordinate systems induced by statistical regularities in scientific language.

2 Representation Layer

2.1 Embedding Model

Let

$$\mathcal{E} : \mathfrak{T} \rightarrow \mathbb{R}^d$$

be a semantic embedding model mapping text segments into a d -dimensional vector space. Here:

- $\mathfrak{T}$ denotes the space of textual expressions,
- $\mathcal{E}$ is a pretrained embedding function,
- $\mathbb{R}^d$ is the embedding coordinate space.

Importantly, $\mathbb{R}^d$ is not assumed to be the ontology of scientific knowledge itself. Rather, it is an empirical representation space induced by the statistical structure of language.

2.2 Discourse Segmentation

For each scientific document i , we extract discourse-role segments:

$$\tau_i^{text} = (t_i^P, t_i^M, t_i^F, t_i^I)$$

where:

- t_i^P is the problem discourse segment,
- t_i^M is the method discourse segment,

- t_i^F is the finding discourse segment,
- t_i^I is the interpretation discourse segment.

These segments are rhetorical or discourse-functional partitions extracted from scientific texts.

2.3 Discourse-Role Embeddings

Applying the embedding model to each segment yields:

$$p_i = \mathcal{E}(t_i^P)$$

$$m_i = \mathcal{E}(t_i^M)$$

$$f_i = \mathcal{E}(t_i^F)$$

$$i_i = \mathcal{E}(t_i^I)$$

with:

$$p_i, m_i, f_i, i_i \in \mathbb{R}^d$$

The tuple:

$$\tau_i = (p_i, m_i, f_i, i_i)$$

defines the embedded *research approach* associated with document i .

3 Empirical State Distributions

We define empirical discourse distributions over discourse-role embeddings:

$$P = \{p_i\}_{i=1}^N$$

$$M = \{m_i\}_{i=1}^N$$

$$F = \{f_i\}_{i=1}^N$$

$$I = \{i_i\}_{i=1}^N$$

These are not assumed to be independent ontological spaces. Instead, they are empirical distributions induced by different discourse functions over the same embedding geometry.

Thus:

$$P, M, F, I \subset \mathbb{R}^d$$

represent semantically conditioned regions of the embedding space.

4 Observed Discourse Trajectories

A scientific document induces an ordered sequence:

$$\tau_i = (p_i, m_i, f_i, i_i)$$

which we interpret empirically as an observed discourse trajectory.

Importantly, this trajectory does not necessarily represent literal motion through an underlying epistemic manifold. Instead, it represents a sequence of semantically related discourse states co-occurring within a scientific reasoning process.

Instead of thinking of the research approach as a single point in the EL, we can interpret τ_i as a trajectory induced by i bridging the spaces P , M , F , and I .

5 Local Displacement Operators

5.1 Definition

For each document i , we define local displacement vectors:

$$\Delta_{PM}^{(i)} = m_i - p_i$$

$$\Delta_{MF}^{(i)} = f_i - m_i$$

$$\Delta_{FI}^{(i)} = i_i - f_i$$

where:

$$\Delta_{PM}^{(i)}, \Delta_{MF}^{(i)}, \Delta_{FI}^{(i)} \in \mathbb{R}^d$$

These vectors are empirical displacement estimates between discourse-role embeddings.

5.2 Interpretation

The displacement vectors are not assumed to be literal epistemic operators. Instead, they are interpreted as:

semantic displacement estimates associated with transitions between discourse roles within scientific documents.

For example:

$$\Delta_{PM}^{(i)}$$

captures the semantic displacement associated with moving from problem discourse to method discourse within document i .

Similarly:

- Δ_{MF} captures transitions from methods to findings,
- Δ_{FI} captures transitions from findings to interpretation.

5.3 Operator Distributions

We define empirical displacement distributions:

$$\Delta_{PM} = \{\Delta_{PM}^{(i)}\}_{i=1}^N$$

$$\Delta_{MF} = \{\Delta_{MF}^{(i)}\}_{i=1}^N$$

$$\Delta_{FI} = \{\Delta_{FI}^{(i)}\}_{i=1}^N$$

These distributions characterize recurrent semantic transition patterns within scientific discourse.

6 Neighborhood Structure

Let:

$$N_k(x)$$

be the k -nearest-neighbor neighborhood of point x under a similarity metric $d(\cdot, \cdot)$. Typically:

$$d(x, y) = 1 - \cos(x, y)$$

or Euclidean distance after normalization.

Neighborhoods define local empirical association structure in embedding space.

7 Conditional Transition Distributions

7.1 Definition

We define conditional transition distributions between discourse-role states.

For $X, Y \in P, M, F, I$:

$$\mathcal{D}(Y \mid x) = \{y_j \mid x_j \in N_k(x)\}, x \in X$$

For example:

$$\mathcal{D}(M \mid p)$$

denotes the empirical distribution of method embeddings associated with problems near p . Similarly:

$$\mathcal{D}(P \mid m) = \{p_j \mid m_j \in N_k(m)\}$$

$$\mathcal{D}(F \mid m) = \{f_j \mid m_j \in N_k(m)\}$$

$$\mathcal{D}(I \mid f) = \{i_j \mid f_j \in N_k(f)\}$$

and so forth.

7.2 Interpretation

These distributions are not displacement vectors. They are conditional empirical distributions over neighboring discourse states.

Thus:

$$\mathcal{D}(Y \mid x)$$

should be interpreted as:

The empirical distribution of discourse states of type Y associated with semantic neighborhoods around state x .

8 Relationship Between Δ and $\mathcal{D}$

The framework distinguishes two fundamentally different empirical objects.

8.1 Observed Local Displacements

$$\Delta_{XY}^{(i)} \in \mathbb{R}^d$$

represents:

a single observed displacement between discourse-role embeddings within document i .

These are trajectory-local quantities.

8.2 Conditional Transition Structure

$$\mathcal{D}(Y \mid x)$$

represents:

The empirical distribution of possible neighboring discourse states associated with semantic region x .

These are population-level conditional structures.

9 Hypothesis Structure

The framework proposes three progressively stronger empirical hypotheses.

- H1: local discourse displacements exhibit non-random structure,
- H2: conditional transition distributions exhibit organized local geometry,
- H3: transition structure possesses predictive value for future semantic occupation patterns.

These hypotheses correspond respectively to:

- (i) existence of structured transition dynamics,
- (ii) organization of local transition landscapes,
- (iii) temporal predictive utility.

10 H1 — Discourse Simplex Structure Hypothesis

10.1 Informal Statement

Scientific papers exhibit coherent discourse-simplex structure. The geometric relationships connecting problems, methods, findings, and interpretations differ systematically from those obtained by randomly reassigning discourse roles across papers.

10.2 Observed Transition Operators

For each paper (i), let:

$$\sigma_i = (p_i, m_i, f_i, i_i)$$

denote its discourse simplex, where:

$$p_i, m_i, f_i, i_i \in \mathbb{R}^d$$

are the corresponding problem, method, finding, and interpretation discourse embeddings. We define three empirical transition operators:

$$\Delta_{PM}^{(i)} = m_i - p_i$$

$$\Delta_{MF}^{(i)} = f_i - m_i$$

$$\Delta_{FI}^{(i)} = i_i - f_i.$$

These vectors define the sequential discourse trajectory:

$$p \rightarrow m \rightarrow f \rightarrow i.$$

10.3 Trajectory and Simplex Geometry

The discourse simplex can be interpreted either as:

- (a) a trajectory composed of three sequential transitions, or
- (b) a 3-simplex whose vertices are

$$(p, m, f, i).$$

The remaining simplex edges are:

$$\Delta_{PF} = \Delta_{PM} + \Delta_{MF}$$

$$\Delta_{MI} = \Delta_{MF} + \Delta_{FI}$$

$$\Delta_{PI} = \Delta_{PM} + \Delta_{MF} + \Delta_{FI}.$$

Therefore the complete tetrahedral geometry is fully determined by the three sequential transition operators.

Testing trajectory structure is thus equivalent to testing simplex structure.

10.4 Feature Representation

For each paper, we compute six geometric features:

Transition Magnitudes

$$|\Delta_{PM}|, \quad |\Delta_{MF}|, \quad |\Delta_{FI}|$$

capturing the semantic displacement associated with each transition.

Transition Alignment

$$\cos(\Delta_{PM}, \Delta_{MF})$$

$$\cos(\Delta_{PM}, \Delta_{FI})$$

$$\cos(\Delta_{MF}, \Delta_{FI})$$

capturing directional relationships between transitions.

Together these define a six-dimensional simplex descriptor:

$$F_i = \left(|\Delta_{PM}|, |\Delta_{MF}|, |\Delta_{FI}|, \cos(\Delta_{PM}, \Delta_{MF}), \cos(\Delta_{PM}, \Delta_{FI}), \cos(\Delta_{MF}, \Delta_{FI}) \right).$$

Collecting all papers yields:

$$F_{obs} = F_{i=1}^N \subset \mathbb{R}^6.$$

10.5 Null Model

A randomized corpus is generated by independently permuting discourse-role assignments:

$$P, \quad M, \quad F, \quad I.$$

Formally:

$$P' = \pi_P(P), \quad M' = \pi_M(M), \quad F' = \pi_F(F), \quad I' = \pi_I(I).$$

This preserves:

- the marginal distribution of each discourse space,
- the geometry of each discourse space,
- corpus-level semantic structure,

while destroying paper-level coupling between discourse roles.

The same six-dimensional feature representation is then computed for the randomized corpus:

$$F_{null}.$$

10.6 Formal Hypothesis

Let:

$$F_{obs}$$

denote the observed simplex-feature distribution and:

$$F_{null}$$

the corresponding distribution under discourse-role randomization.
H1 states that:

$$F_{obs} \neq F_{null}.$$

Equivalently, the joint geometry of discourse simplices cannot be explained solely by the marginal structure of the individual discourse spaces.

10.7 Primary Statistical Test

We compare the observed and randomized feature distributions using the Energy Distance:

$$D_E^2(F_{obs}, F_{null}) = 2, \mathbb{E}|x - y| \mathbb{E}|x - x'| \mathbb{E}|y - y'|,$$

where:

$$x, x' \sim F_{obs}$$

and

$$y, y' \sim F_{null}.$$

Energy Distance satisfies:

$$D_E^2 = 0$$

if and only if the two distributions are identical.

Positive values indicate differences in any aspect of the multivariate geometry, including location, scale, and correlation structure.

10.8 Permutation Significance

Statistical significance is estimated through repeated random discourse-role permutations.

For each permutation (b):

- (a) discourse roles are independently permuted,
- (b) a randomized feature distribution ($F_{null}^{(b)}$) is generated,
- (c) the Energy Distance

$$D_E^{2(b)}$$

is computed.

The empirical p-value is:

$$p = \frac{1 + \sum_{b=1}^B \mathbf{1} \left[D_E^{2(b)} \geq D_{E,obs}^2 \right]}{B + 1}.$$

The null hypothesis is:

$$H_0 : F_{obs} = F_{null}.$$

The alternative hypothesis is:

$$H_A : F_{obs} \neq F_{null}.$$

10.9 Decision Rule

H1 is considered supported when:

$$p < 0.05.$$

Support for H1 indicates that scientific discourse exhibits coherent simplex structure beyond that expected from independently assembled discourse roles.

10.10 Secondary Diagnostics

To facilitate interpretation of the multivariate result, we additionally compute:

- (a) Wasserstein effect sizes for each of the six simplex features,
- (b) two-sample Kolmogorov–Smirnov statistics for each feature individually,
- (c) the six-edge simplex profile

$$(|\Delta_{PM}|, |\Delta_{MF}|, |\Delta_{FI}|, |\Delta_{PF}|, |\Delta_{PI}|, |\Delta_{MI}|)$$

used for visualization and exploratory analysis.

These quantities are descriptive and do not constitute the primary hypothesis test.

11 H2 — Conditional Transition Organization Hypothesis

11.1 Informal Statement

Conditional transition operators exhibit organized local structure. Papers that are semantically nearby in one discourse space tend to occupy systematically related regions in other discourse spaces, beyond what would be expected by chance.

11.2 Formal Statement

Let:

$$\mathcal{D}(Y \mid x)$$

denote a conditional transition operator relating a source discourse state ($x \in X$) to neighboring states in discourse space (Y).

H2 states that the local organization induced by these operators differs significantly from randomized conditional associations.

11.3 Observed Transition Structure

For each ordered pair of discourse spaces:

$$X, Y \in P, M, F, I, \quad X \neq Y,$$

we estimate a conditional transition operator:

$$\mathcal{D}(Y \mid x).$$

The analysis is performed for all twelve directional transitions:

$$P \rightarrow M, P \rightarrow F, P \rightarrow I,$$

$$M \rightarrow P, M \rightarrow F, M \rightarrow I,$$

$$F \rightarrow P, F \rightarrow M, F \rightarrow I,$$

$$I \rightarrow P, I \rightarrow M, I \rightarrow F.$$

For a query state (x), the (k)-nearest neighbors are identified in the source space (X).

The corresponding target embeddings in (Y) define an empirical transition neighborhood:

$$Y_{\text{nbr}}.$$

A random sample of equal size drawn from (Y) defines:

$$Y_{\text{rand}}.$$

The local transition organization is quantified by the Wasserstein distance:

$$W_{\text{obs}} = W(Y_{\text{nbr}}, Y_{\text{rand}}).$$

Large values indicate that target states associated with neighboring source states differ substantially from random target samples.

11.4 Null Model

The null hypothesis assumes that source-space neighborhoods contain no information about target-space organization.

Under this model, transition structure is generated entirely from random samples of the target space.

A null distribution is estimated by repeatedly drawing independent random samples:

$$Y^{(1)} * \text{rand}, Y^{(2)} * \text{rand} \subset Y$$

and computing:

$$W_{\text{null}} = W\left(Y^{(1)} * \text{rand}, Y^{(2)} * \text{rand}\right).$$

Repeating this procedure produces a null pool:

$$W_{\text{null}}^{(1)}, \dots, W_{\text{null}}^{(N)}.$$

This preserves the geometry of the target space while removing conditional organization.

11.5 Bootstrap Significance Test

For each transition operator, a bootstrap distribution is constructed by repeatedly sampling with replacement from the null pool and computing the mean Wasserstein distance.

Let:

$$\bar{W}_{null}^{(b)}$$

denote the bootstrap mean obtained in iteration (b).

The empirical p-value is:

$$p = \frac{1 + \sum_{b=1}^B \mathbf{1} \left[\bar{W} * null^{(b)} \geq W * obs \right]}{B + 1}.$$

The null hypothesis is:

$$H_0 : W_{obs} \sim W_{null}.$$

The alternative hypothesis is:

$$H_A : W_{obs} > W_{null}.$$

Rejecting (H_0) indicates that source-space neighborhoods induce non-random organization in the target space.

11.6 Effect Size

In addition to significance testing, we compute a standardized effect size:

$$z = \frac{W_{obs} - \mu_{null}}{\sigma_{null}},$$

where:

$$\mu_{null}$$

and

$$\sigma_{null}$$

are the mean and standard deviation of the bootstrap null distribution.

The z-score quantifies the strength of local transition organization relative to the randomized baseline.

11.7 Decision Rule

H2 is considered supported when at least three of the twelve directional transition operators satisfy:

$$p < 0.05.$$

Support for H2 indicates that conditional transition operators capture meaningful local organization linking discourse spaces.

11.8 Directional Asymmetry Analysis

As an additional characterization of the transition structure, we evaluate directional asymmetry between pairs of discourse spaces.

For each unordered pair:

$$(X, Y),$$

we estimate an effective branching factor in both directions:

$$X \rightarrow Y$$

and

$$Y \rightarrow X.$$

Branching is quantified through the entropy of target-space neighborhoods, yielding an effective number of reachable semantic regions.

Observed directional differences are compared against a permutation null model obtained by randomly exchanging source and target assignments.

Significant asymmetry indicates that:

$$\mathcal{D}(Y|x) \neq \mathcal{D}(X|y),$$

revealing direction-dependent transition structure.

Unlike H2 itself, asymmetry is interpreted as a descriptive property of the transition operators rather than a prerequisite for their existence.

12 H3 — Predictive Transition Hypothesis

12.1 Informal Statement

Historical discourse-transition structure contains predictive information about future problem-space occupation.

Specifically, a transition model learned from historical discourse embeddings should predict future problem embeddings more accurately than a persistence baseline.

12.2 Temporal Train/Test Construction

Let the corpus be partitioned at a temporal cutoff t .

Historical papers define:

$$P_{hist}, M_{hist}, F_{hist}, I_{hist}$$

while papers published after t define:

$$P_{fut}, M_{fut}, F_{fut}, I_{fut}.$$

Only information available before t is used for model estimation.

12.3 Learning a Transition Operator

A predictive transition operator is estimated from historical data by fitting a multi-output Ridge regression:

$$\hat{\mathcal{R}} : I \rightarrow P$$

such that

$$P_{hist} \approx \hat{\mathcal{R}}(I_{hist}).$$

Operationally, this corresponds to learning a linear map:

$$W$$

minimizing:

$$\|P_{hist} - I_{hist}W\|^2 + \alpha\|W\|^2.$$

Future problem embeddings are then predicted as:

$$\hat{P}_{fut} = \hat{\mathcal{R}}(I_{fut}).$$

This model represents an empirical approximation of the theoretical interpretation-to-problem transition.

12.4 Persistence Baseline

As a baseline predictor, we consider a persistence model that assumes no temporal change.

Its prediction error is defined as the Wasserstein distance between historical and future problem distributions:

$$W_{baseline} = W(P_{hist}, P_{fut}).$$

The EDEL prediction error is:

$$W_{EDEL} = W(\hat{P}_{fut}, P_{fut}).$$

Both distances are computed directly in problem-embedding space using optimal transport with Euclidean ground cost.

12.5 Predictive Gain

Forecasting performance is evaluated through predictive gain:

$$G = W_{baseline} - W_{EDEL}.$$

Positive values indicate that the transition model predicts future problem states more accurately than persistence:

$$G > 0.$$

The primary prediction of H3 is:

$$G > 0.$$

12.6 Predictive Gain Significance

Statistical significance is estimated through temporal permutation.

For each permutation b :

- (a) publication years are randomly reassigned,
- (b) historical and future partitions are reconstructed,
- (c) the transition model is re-estimated,
- (d) predictive gain $G^{(b)}$ is recomputed.

This produces a null distribution:

$$\{G^{(1)}, \dots, G^{(B)}\}.$$

The p-value is estimated as:

$$p_G = \frac{1 + \sum_{b=1}^B \mathbf{1}[G^{(b)} \geq G_{obs}]}{B + 1}.$$

H3 is supported when:

$$G_{obs} > 0$$

and

$$p_G < 0.05.$$

12.7 Spatial Predictive Alignment

As a secondary diagnostic, we evaluate whether predicted semantic expansion occurs in the same regions that subsequently experience empirical growth.

Problem embeddings are partitioned into K semantic regions using KMeans clustering.

Let:

$$c_{hist}, \quad c_{pred}, \quad c_{fut}$$

denote cluster occupancy counts for historical, predicted, and future papers respectively.

Predicted and observed density changes are defined as:

$$x_i = c_{pred,i} - c_{hist,i}$$

and

$$y_i = c_{fut,i} - c_{hist,i}.$$

Bivariate Moran's I is then computed:

$$I_B = \frac{n}{W} \frac{\sum_i \sum_j w_{ij} (x_i - \bar{x})(y_j - \bar{y})}{\sum_i (x_i - \bar{x})^2}$$

where w_{ij} is a row-standardized inverse-distance weight matrix defined over cluster centroids.

Positive values indicate that regions predicted to expand tend to coincide with regions that actually expand.

12.8 Spatial Significance

A second temporal permutation procedure is used to evaluate spatial alignment.

For each permutation:

- (a) publication years are shuffled,
- (b) historical and future partitions are reconstructed,
- (c) clustering and density estimation are repeated,
- (d) a new Moran statistic $I_B^{(b)}$ is computed.

The resulting p-value is:

$$p_I = \frac{1 + \sum_{b=1}^B \mathbf{1} \left[I_B^{(b)} \geq I_{obs} \right]}{B + 1}.$$

Unlike predictive gain, Bivariate Moran's I is interpreted as an exploratory measure of spatial predictive alignment rather than the primary hypothesis test.

13 Novelty and Closure Metrics

13.1 Cycle Closure Approximation

A complete epistemic cycle may be interpreted theoretically as a transformation:

$$\Phi : P \rightarrow P$$

mapping an initial problem state into a future problem state generated through scientific reasoning.

Empirically, however, future problems are not directly observed within a single scientific abstract. Therefore, we define a first-order empirical approximation based on the semantic displacement between the initial problem embedding and the interpretation embedding.

For document i :

$$\Delta_{cycle}^{(i)} = i_i - p_i$$

Since:

$$(i_i - f_i) + (f_i - m_i) + (m_i - p_i) = i_i - p_i$$

this quantity represents the net open-cycle semantic displacement accumulated along the observed discourse trajectory.

13.2 Closure Condition

We define an approximate closure condition:

$$\|i_i - p_i\| < \epsilon$$

for some small threshold ϵ .

Intuitively, this corresponds to trajectories in which the interpretation semantically returns near the original problem region.

Such trajectories may correspond to:

- confirmatory studies,

- replication experiments,
- local theoretical validation,
- or highly incremental scientific progress.

In contrast:

$$||i_i - p_i|| \gg 0$$

indicates trajectories whose interpretations diverge substantially from the original problem framing.

These may correspond to:

- conceptual expansion,
- methodological transfer,
- paradigm generalization,
- or broader epistemic reframing.

13.3 Novelty Metric

We define the empirical novelty score:

$$\mathcal{N}(i) = ||i_i - p_i||$$

which measures the semantic displacement between the initial problem framing and the final interpretation framing of a scientific trajectory.

Importantly, this metric does not measure truth, correctness, or scientific quality. Instead, it approximates the degree of semantic and conceptual displacement induced by the discourse trajectory.

13.4 Future Problem Approximation

A more complete approximation of epistemic closure may be constructed using conditional transition distributions.

Let:

$$\mathcal{D}(P \mid i)$$

be the empirical distribution of problem embeddings associated with semantic neighborhoods around interpretation state i .

We define the expected future problem embedding:

$$\bar{p}_i = \mathbb{E}[P \mid i]$$

which may be estimated operationally using nearest-neighbor averaging over problem embeddings associated with neighboring interpretation states.

This yields an approximate net epistemic displacement:

$$\Delta_E^{(i)} = \bar{p}_i - p_i$$

where:

$$||\Delta_E^{(i)}||$$

represents an estimate of the future problem displacement induced by the discourse trajectory.

14 Theoretical Interpretation Layer

The empirical framework described above constructs semantic representations from scientific discourse without requiring that embedding geometry correspond directly to literal epistemic structure. However, the observed regularities motivate a higher-level theoretical interpretation in which scientific activity is represented as a temporally evolving higher-order epistemic structure.

Importantly, EDEL distinguishes between:

- (a) an *empirical ontology*, in which semantic and topological structures are reconstructed from scientific discourse data, and
- (b) a *generative ontology*, in which these structures become operational substrates for synthetic epistemic agents capable of generating and composing scientific reasoning trajectories.

14.1 EDEL as a Layered Formal Structure

We define empirical EDEL as the pair:

$$\hat{\mathcal{E}} = (\hat{\mathcal{E}}, \hat{\mathcal{T}})$$

where:

- $\hat{\mathcal{E}}$ defines the semantic embedding substrate,
- $\hat{\mathcal{T}}$ defines the temporally evolving epistemic-topological structure reconstructed from discourse.

The semantic embedding is defined as the function:

$$\hat{\mathcal{E}} : \mathfrak{T} \rightarrow E$$

where:

- $\mathbb{R}^d$ is the semantic embedding space,
- $\mathfrak{T}$ is the space of all possible texts

Given a scientific artifact a_i , segmentation produces discourse-role embeddings:

$$p_i, m_i, f_i, i_i \in E$$

corresponding respectively to problem, method, finding, and interpretation discourse.

Importantly, these embeddings are treated as empirical semantic representations rather than direct epistemic states.

14.2 Discourse Simplices

Each research approach induces a discourse simplex:

$$\sigma_i = (p_i, m_i, f_i, i_i)$$

interpreted geometrically as a 3-simplex embedded in semantic space.

Within this interpretation:

- vertices correspond to discourse-role embeddings,

- edges correspond to pairwise discourse transitions,
- triangular faces correspond to partial epistemic configurations,
- the tetrahedral simplex represents a complete discourse trajectory associated with a research approach.

Unlike conventional graph-based representations, discourse simplices preserve higher-order relational structure between all discourse components simultaneously.

14.3 Epistemic Simplicial Complex

The global epistemic organization is represented as a temporally indexed directed simplicial complex:

$$\mathcal{T} = \{\mathcal{K}_t\}_{t \in \mathbb{T}}$$

where each temporal epistemic complex:

$$\mathcal{K}_t = (\mathcal{S}_t, \mathcal{G}_t)$$

contains:

- a set of discourse simplices:

$$\mathcal{S}_t = \{\sigma_i\}$$

where each:

$$\sigma_i = (p_i, m_i, f_i, i_i)$$

represents the discourse embedding tuple associated with a scientific work;

- a generalized simplicial gluing operator:

$$\mathcal{G}_t : \mathcal{S}_t \rightarrow 2^{\mathcal{S}_t}$$

defining semantic attachment relations between discourse simplices.

The gluing operator is induced by a family of empirical transition distributions:

$$\mathcal{D}_{X \rightarrow Y}(Y \mid x)$$

defined over discourse spaces:

$$X, Y \in \{P, M, F, I\}$$

with:

$$x \in X$$

Each transition distribution characterizes empirical semantic neighborhoods across discourse roles.

For example:

$$\mathcal{D}_{P \rightarrow M}(M \mid p)$$

describes methods empirically associated with neighboring problem states, while:

$$\mathcal{D}_{M \rightarrow P}(P \mid m)$$

describes problem regions empirically associated with neighboring methods.
Importantly:

$$\mathcal{D}_{P \rightarrow M}(M \mid p) \neq \mathcal{D}_{M \rightarrow P}(P \mid m)$$

in general.

The transition structure is therefore intrinsically asymmetric.

The generalized gluing operator aggregates these local transition relations into a higher-order epistemic topology.

Formally:

$$\sigma_j \in \mathcal{G}_t(\sigma_i)$$

iff there exists at least one admissible discourse transition relation:

$$x_j \in \mathcal{D}_{X \rightarrow Y}(Y \mid x_i)$$

for some:

$$X, Y \in \{P, M, F, I\}$$

with:

$$x_i \in \sigma_i \quad \text{and} \quad x_j \in \sigma_j$$

The resulting structure defines a directed higher-order epistemic simplicial complex in which discourse simplices become connected through asymmetric semantic transition relations between epistemic aspects.

A 3-simplex σ_i shares a 1-simplex $x \in \sigma_i$ with all neighboring simplices $\sigma_j \in \mathcal{G}(\sigma_i)$.

14.4 Empirical Interpretation

Within the empirical ontology, the simplicial complex is interpreted as a statistical reconstruction of scientific discourse organization.

In this setting:

- semantic geometry approximates discourse similarity,
- simplicial attachment approximates higher-order epistemic organization,
- transition operators approximate empirical semantic affordances between discourse regions.

Importantly, the framework does not claim that embeddings or simplices constitute literal epistemic states. Rather, they provide computable approximations of structural regularities observable in scientific discourse corpora.

14.5 Closure and Epistemic Expansion

The displacement:

$$\Delta_E(i) = i_i - p_i$$

defines the net semantic displacement between the initial problem discourse and the final interpretive discourse of a scientific artifact.

Its norm:

$$\mathcal{N}(i) = \|i_i - p_i\|$$

is interpreted as an empirical approximation of epistemic expansion.

Small values correspond to semantically closed trajectories in which interpretive discourse remains close to the original problem framing, while large values correspond to semantically expansive trajectories generating displaced conceptual regions.

Higher-order geometric observables may also be defined over discourse simplices, including:

- edge lengths,
- face areas,
- simplex volumes,
- operator-space simplices constructed from displacement operators.

These quantities characterize increasingly complex forms of epistemic transformation geometry.

15 Generative Interpretation Layer

We define generative EDEL as:

$$\mathfrak{E} = (\mathcal{E}, \mathcal{T}, \mathcal{C})$$

where:

$$\mathcal{C} = (\mathcal{O}, \mathcal{M}, \circ)$$

defines a compositional epistemic structure.

Here:

- $\mathcal{O} = \{P, M, F, I\}$ defines discourse-role objects,
- $\mathcal{M}$ defines epistemic morphisms between discourse spaces,
- $\circ$ defines compositional structure over epistemic transformations.

In this interpretation:

$$P \rightarrow M \rightarrow F \rightarrow I$$

is treated as a compositional scientific reasoning trajectory rather than merely an observed semantic sequence.

The conditional transition operators become generative affordance structures.

For example:

$$m^* \sim \mathcal{D}(M \mid p)$$

defines admissible methodological continuations for a problem state p .
Similarly:

$$f^* \sim \mathcal{D}(F \mid m^*)$$

and:

$$i^* \sim \mathcal{D}(I \mid f^*)$$

define generative transitions through epistemic discourse spaces.

Agents may therefore construct novel discourse simplices not explicitly observed in the empirical corpus:

$$\sigma^* = (p, m^*, f^*, i^*)$$

which can subsequently attach to the evolving epistemic complex.

Within this ontology, scientific discovery is interpreted as growth and restructuring of the epistemic simplicial complex through compositional epistemic operations.

Importantly, the category-theoretic layer is not directly observable empirically. Rather, it provides a formal operational semantics for generative scientific systems built on top of empirically reconstructed epistemic topology.

15.1 Epistemic Value

The original Epistemic Landscape (EL) framework models scientific exploration as search over a landscape whose topology represents research approaches and whose height corresponds to their epistemic value. In EDEL, this idea is generalized from isolated points to higher-order epistemic structures represented as discourse simplices.

Formally, let:

$$\sigma_i = (p_i, m_i, f_i, i_i)$$

denote the discourse simplex associated with scientific artifact i , where:

- p_i is the problem discourse embedding,
- m_i is the method discourse embedding,
- f_i is the finding discourse embedding,
- i_i is the interpretation discourse embedding.

We define the epistemic value operator:

$$\mathcal{V} : \mathcal{S} \rightarrow \mathbb{R}$$

where:

$$\mathcal{S} = \{\sigma_i\}$$

is the set of discourse simplices composing the EDEL complex.

The quantity:

$$\mathcal{V}(\sigma_i)$$

represents the epistemic utility associated with the corresponding scientific reasoning configuration. Importantly, epistemic value is treated as formally distinct from semantic geometry and topological structure. A simplex may occupy a dense or central region of the epistemic complex while still having low epistemic utility, and conversely highly valuable simplices may emerge in sparse or weakly connected regions.

This distinction separates:

- the *structure* of scientific discourse,
- from the *value* of exploring particular epistemic regions.

Empirical EDEL In empirical EDEL, epistemic value cannot be observed directly and must therefore be approximated through external scientometric or social indicators. Let:

$$\hat{\mathcal{V}}(\sigma_i)$$

denote an empirical estimator of epistemic value.

Examples include:

- citation counts,
- field-normalized citations,
- PageRank-style influence metrics,
- theorem reuse frequency,
- downstream clinical or technological adoption.

In the present work, we primarily consider:

$$\hat{\mathcal{V}}(\sigma_i) = \log(1 + c_i)$$

where c_i denotes the citation count associated with artifact i .

These quantities should not be interpreted as direct measures of truth or intrinsic scientific merit. Instead, they provide imperfect observational proxies for the social and epistemic impact of scientific artifacts. Epistemic value in empirical EDEL therefore remains partially latent and institutionally mediated.

Generative EDEL In generative EDEL, epistemic value is not inferred from historical traces but defined explicitly through the ground-truth structure of the target problem domain.

Let:

$$\mathfrak{G}$$

denote the ground-truth environment associated with a scientific domain or task. Epistemic value is then defined as:

$$\mathcal{V} : \mathcal{S} \times \mathfrak{G} \rightarrow \mathbb{R}$$

assigning utility to discourse simplices according to their contribution toward solving or explaining the target environment.

Examples include:

- predictive accuracy,

- theorem proving success,
- simulation performance,
- optimization objectives,
- explanatory compression,
- environment reward.

In this setting, epistemic value becomes operationally well-defined and externally grounded. Scientific agents operating over the EDEL structure can therefore optimize exploration policies relative to explicit utility functions while remaining constrained by the transition structure of the epistemic complex.

Extended EDEL Formalism With the inclusion of epistemic value, the empirical EDEL framework can be summarized as:

$$\hat{\mathfrak{E}} = (\hat{\mathcal{E}}, \hat{\mathcal{T}}, \hat{\mathcal{V}})$$

where:

- $\hat{\mathcal{E}}$ defines the semantic and topological epistemic structure,
- $\hat{\mathcal{T}}$ defines the epistemic transition dynamics,
- $\hat{\mathcal{V}}$ defines the epistemic valuation structure.

Including the compositional semantics layer on the generative ontology yields the extended formulation:

$$\mathfrak{E} = (\mathcal{E}, \mathcal{T}, \mathcal{C}, \mathcal{V})$$

where:

$$\mathcal{C}$$

defines the compositional and categorical structure governing admissible epistemic operations and scientific reasoning processes.